

Lab 3 – Architecture Selection & Justification

System: Peer-to-Peer Carpooling & Expense Splitter

Architecture Selection: We chose Layered Architecture for the Peer-to-Peer Carpooling & Expense Splitter System.

Architectural Choice

The system is organized into Presentation, Business and Data layers. The Presentation Layer contains the Carpool UI for Rider Commuters and Driver Hosts. The Business Layer contains the Ride Manager, Route Matching Service and Payment Service. The Data Layer contains the Carpool Database for storing user, ride, route and travel-history information.

Reason 1

Layered Architecture provides clear separation of responsibilities. User interaction is handled separately from core operations such as ride management, route compatibility checking and expense splitting. This makes the system easier to understand, maintain and modify.

Reason 2

The architecture is suitable for this system because the main functions can be separated into focused components while still working together through defined interfaces. The Route Matching Service can handle overlapping routes and the 1.5 km detour tolerance, while the Payment Service handles automated fuel-cost sharing.

Security Advantage

Sensitive information can be isolated within the Data Layer and accessed through controlled application components rather than directly through the user interface. This is useful for protecting user ratings, driving-license verification documents and travel histories, which the system is required to store securely.

Performance Benefit

Separating responsibilities allows each layer to perform only the processing it requires. Route compatibility calculations can be handled by the Route Matching Service, while the Carpool UI remains focused on user interactions. This helps keep the user experience responsive and avoids unnecessary processing in the presentation layer.